Lesson 9-4 · Session 1

Substitute and Solve

Name _____ Date _____ Period _____

Maryland Standard 6.AT.11 · Learning target — I can substitute a value into an equation to find the matching output.

Show a model, calculation, or explanation. Answer every part and label units when needed.

Exercises 1–12 earn 1 point for a correct response and 1 for supporting work. Spiral Review and Test Practice earn 1 point each.

Learn from a model

REMEMBER

- To find y for a given x (or x for a given y), substitute the known value into the equation and solve.
- Follow order of operations carefully after substituting.
- Watch out — substituting into an equation with a fee needs the order of operations.
- Check it — your answer should grow when the input grows, unless the rule subtracts.

WORKED EXAMPLE

Substituting to Find the Output A press makes 22 posters each hour. Write an equation, then find the posters made in 15 hours.

Step 1 Define: h = hours running, p = posters made. Posters depend on hours.

Step 2 The rate is 22 posters for each hour, so the rule is p = 22h.

Step 3 Substitute 15 for h: 22 × 15 330 posters.

Step 4 Check the context: many more hours gave a much larger count, so this fits.

Answer p = 22h, so a 15-hour run makes 330 posters.

Practice

1. Using y = 3x + 2, find y when x = 5.

Answer _____

SHOW YOUR THINKING

2. Using y = 4x − 1, find x when y = 15. (Hint: substitute, then solve the equation.)

Answer _____

SHOW YOUR THINKING

3. Using y = 2x + 6, find x when y = 20.

Answer _____

SHOW YOUR THINKING

4. Substitute the given value of x and find y.

a. y = 5x + 3, when x = 4 y = _____
b. y = 7x, when x = 9 y = _____
c. y = 2x + 11, when x = 8 y = _____
d. y = 6x − 5, when x = 10 y = _____
e. y = x + 19, when x = 12 y = _____

Answer _____

SHOW YOUR THINKING

5. Sprinkler. A sprinkler puts out 17 gallons of water each hour it runs. Write an equation for the gallons g used in h hours, then find the gallons used in 24 hours.

Answer _____

SHOW YOUR THINKING

6. Carnival Tickets. A carnival charges $9 for each ride ticket plus a $5 entry fee. Write an equation for the total cost y for x tickets, then find the cost of 12 tickets.

Answer _____

SHOW YOUR THINKING

7. Stocking Shelves. A worker uses the rule b = 16m to find the boxes b unpacked in m minutes. Find the boxes unpacked in 45 minutes, then explain whether your answer is reasonable.

Answer _____

EXPLAIN WITH MATHEMATICAL EVIDENCE

8. Using y = 8x + 6, Selena said that when x = 7 the value of y is 98. She added 6 to 8 first and then multiplied by 7. Explain her mistake, then find the correct value of y.

Answer _____

EXPLAIN WITH MATHEMATICAL EVIDENCE

9. For y = 12x, find y when x = 7.

Answer _____ value

SHOW YOUR THINKING

10. For c = 8h + 15, find c when h = 5.

Answer _____ dollars

SHOW YOUR THINKING

11. Two Inputs. For y = 4x + 9, find the output when x = 6 and when x = 12.

Answer _____ both values

SHOW YOUR THINKING

12. Find the Mistake. For c = 6h + 10 with h = 3, Lu added first and wrote 78. Explain the mistake.

Answer _____ the mistake and the value

EXPLAIN WITH MATHEMATICAL EVIDENCE

Spiral review

13. From Lesson 9-3 Session 2. A $40 fee plus $9 an hour gives what equation?

Answer _____

SHOW YOUR THINKING

14. From Lesson 8-5 Session 2. Is n = 6 a solution of n ≤ 6?

Answer _____

SHOW YOUR THINKING

15. From Lesson 6-8 Session 2. Factor 8n + 20.

Answer _____

SHOW YOUR THINKING

Test practice

16. For y = 7x, find y when x = 9.

□ A. 16	□ B. 63
□ C. 79	□ D. 0.78

17. For c = 4h + 12, find c when h = 6.

□ A. 22	□ B. 96
□ C. 36	□ D. 72

18. For y = 10x + 5, find y when x = 3.

□ A. 150	□ B. 45
□ C. 18	□ D. 35

Check and revise

CHECK YOUR MATHEMATICS	HOW DID IT GO?
Keep both sides equal and check in the original equation. Every part answered / Work shown / Units labeled when needed	I can do this: Independently / With a hint / With help Question to revisit: _____

One correction or question I want to discuss: _____

Substitute one input-output pair into your rule. Check units and whether your answer fits the situation.